

Augmenting Path and Improving Dual

P-D: f dual feas.

$\tilde{f}$ sol'n to DRD

$$f_{\text{new}} = f + \theta \tilde{f}$$

→ Defn: An aug. path P (given f , feasible flow) is

a path from s to t in the undirected graph

st. an edge (i, j) used by P in forward dir: $f_{ij} < b_{ij}$

" " backward $f_{ij} > 0$

Can use P to augment flow by:

$$\Delta = \min_{(i,j) \in P} \begin{cases} b_{ij} - f_{ij} \\ f_{ij} \end{cases} \quad \forall (i,j) \text{ forward backward.}$$

Labeling Algorithm: Finding the AP

Will start from node s
with label (see next slide), and propagate
labels. "Propagating" - also called "scanning".

Move on until no more nodes
can be labelled.

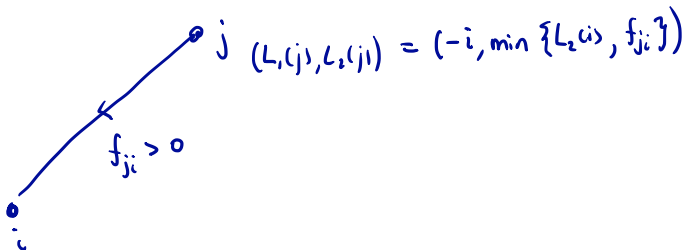
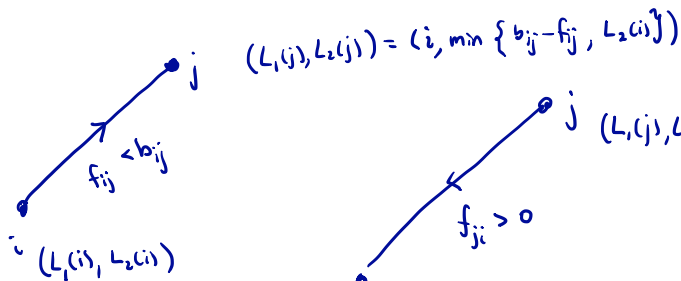
Assigning Labels

Each node has label with 2 parts:

Node i : $(L_1(i), L_2(i))$

$L_1(i)$: "from where i was labelled"

$L_2(i)$: amount of extra flow that can be brought from s to i .



Labeling Algorithm

Maintain list of labelled but
unscanned nodes. start with $LIST = \{s\}$,

$$L_2(s) = +\infty.$$

Pick nodes from $LIST$, scan, label then remove
the picked node from $LIST$.

STOP when $LIST = \emptyset$

Optimality

Thm: If FF terminates, then
we are at opt. soln.

pf: When FF stops — some nodes are labelled
some are not. $S \subseteq W$ labelled, $\bar{W}$ unlabelled.

Note: $\forall (i, j), i \in W, j \in \bar{W}$ must have: $\underline{f_{ij} = b_{ij}}$
 $i \in \bar{W}, j \in W$ $\underline{f_{ij} = 0}$.

$(W, \bar{W})$ is an optimal, i.e., min cut, hence
Flow f is optimal.

Finiteness of Algorithm

Thm: When $b \in \mathbb{Q}$, FF
terminates.